



A
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on
**Plant Recognition Using Deep Learning and
Machine Learning**
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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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ABSTRACT

Identifying plants correctly is vital for ecological studies, farming, and conservation of the environment. Relying solely on traditional taxonomy—which takes a lot of time and is subject to human error because it is based on manual observation of the morphological traits— is impractical. In this paper, we propose an intelligent, vision-based automated taxonomy system to identify plants using leaves and to improve results through generative AI.

The system employs a fine-tuned ResNet-18 convolutional neural network (CNN) to classify plants based on a diverse dataset on different plant species and their leaf morphology. After identification, the model describes the traits and characteristics of the plant, ecological importance, and, when applicable, medicinal value using Google’s Gemini Generative AI.

The entire framework runs on a Streamlit-based web application where users can upload images and interact dynamically with the AI’s plant knowledge to confirm the model’s prediction. We conducted strong experimental evaluations of the system that confirmed the accuracy of knowledge and the identification of plants.

This framework is the first step toward the use of deep learning and large language models (LLMs) for scalable and interactive plant identification systems aimed at knowledge and ecological education.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
ResNet	Residual Network
API	Application Programming Interface
LLM	Large Language Model
FC	Fully Connected
ReLU	Rectified Linear Unit
GPU	Graphics Processing Unit
RGB	Red Green Blue
CVPR	Conference on Computer Vision and Pattern Recognition
BMVC	British Machine Vision Conference

CHAPTER 1: INTRODUCTION

1.1 Introduction

The foundation of life on earth is laid by plants. They maintain an ecological balance and support their earthly ecosystems. They provide and contribute as building blocks for biodiversity; they regulate the balance of gases in the atmosphere; they contribute to the formation of food and shelter and raw materials; they serve as a source of medicines and industrial raw materials. The presence of plants in the various sectors of life such as agriculture, medicines, the environment, and even the climate makes their identification important. Biodiversity, sustainable agriculture, and conservation of ecological balance are crucial to knowing the existence and presence of various plant species.

Plant identification has been undertaken traditionally through taxonomic methods based on the morphological features such as the shape of the leaf, venation, structure of the flower and other structures, and the flower's reproductive organs. Although completely scientific, methods of taxonomic plant identification are time-consuming and require a plant specialist. In the case of the larger scales or real world such as biodiversity surveys and agriculture monitoring activities, manual identification becomes ineffective and human error takes over.

Recently, automated plant identification has been brought about by rapid leaps in human and machine intelligence (AI) and computer vision methods based on

human intelligence. These systems automate the identification of plant components by visual analysis of the leaves, which is the most accessible and most identifying part of the plants. Of the various AI systems, those based on deep learning and in particular Convolutional Neural Networks (CNNs) are showing very promising results in image classification.

In image classification of various plant species, CNN architectures such as ResNet, VGG, and EfficientNet have proven very important by bringing inspiration and keeping a balance between the various elements of this classification and enabling machines to learn hierarchical feature representations directly from raw image data and classifying. They provide automated plant identification systems with the capacity to detect multiple plant features (Plant Attributes) which are the basis for classification, and also other features such as texture, edges, and color proxies.

and venation patterns for plant species recognition. Nevertheless, the majority of the AI-assisted systems, despite their high accuracy, only identified plant species and did little in the way of providing species identification, contextual knowledge or species classification explanations.

There is an evident knowledge prediction and understanding of the identified species. In reality, he or she would like to understand the plant identification, the plant species properties, the plant identification ecological importance, and the plant species identification useful applications. Because of the above knowledge prediction and understanding, the potential of generative AI for plant species identification along with plant species classification systems has been considered. A generative AI model can produce human-like text and therefore, plant identification and classification systems can be viewed as intelligent assistants systems.

The model of the classification combined a Fine-tune model of ResNet-18 for robust plant identification and classification, along with a generative AI model for the classification and identification. The model of the classification leverages both systems and identifies the plant species and the image of the plant leaf, along with an explanation of the botanical, ecological, and medicinal uses of the identified plant.

The classification and identification have a dynamically deployed Streamlit-based web application for the classification, and its classification and identification therefore, the system for the which the user has conceptual proposes, along with system of plant species identification and classification synthesis, along with the system for plant species identification an AI authenticated and classification, is of an exceedingly high accuracy. The classification system provides a robust, reliable, plant species classification and identification system which uses a generative AI model alongside a Fine-tune model of framework as well as industrial identification.

The classification and identification intelligent classification systems aim to educate people within the ecological and welfare preservation. In addition, the classification and identification intelligent classification systems of plant species aim to provide a robust and reliable system for the classification and identification of the plant species along with the system of the mingapore. In addition to the identification of the intelligent systems, the classification and identification systems of the intelligent systems along with the plant classification systems aim to preserve and sustain the areas of wide ecological and environmental systems of plant species identification and classification systems, the preservation and sustainable systems of plant species identification classification along with the systems of plant species preservation and farming.

1.2 Project Description

This project is geared towards developing an intelligent plant identification system. It combines deep learning image classification with generative AI-powered knowledge generation. The system is capable of identifying various species of plants using leaf images and describes the identified plant in a detailed, user-friendly manner.

Within the scope of this project, the following primary AI components are explored:

Deep Learning-Based Image Classification

A Convolutional Neural Network (CNN) model of ResNet-18 is used to classify the species of the plants. The model is developed and trained on a dataset consisting of images of various plant leaf species. Initially, ResNet-18 model is trained on a large dataset of images from a totally different domain. As it has learned various aspects of classification from the general images, the model is adapted using this dataset of leaf images to help with the plant classification.

Generative AI Integration

In order to give the system more capabilities, a model of Google Gemini is also integrated. After the system performs classification, the predicted name of the plant becomes an input to the model for generation of summative content describing the plant. This summative content may describe the plant's ecosystem, its ecology and/or medicinal uses, as well as its habitat and its structure.

The system's workflow is kept simple and intuitive. A user uploads a leaf image through the interface, after which the leaf image undergoes pre-processing and is

then analyzed by the system through the CNN model. The model creates the prediction of the species and then returns the result to the generative AI, which, in turn, provides a detailed appeared summative content describing the various aspects of the plant. In the final system, the output result is accompanied by the input image uploaded by the user, the name of the predicted species and the summative content describing the plant species.

The system uses Streamlit and is designed to be user-friendly and interactive. It allows for predictions in real-time and makes no demands on the user to possess technical computer skills.

The project has the following features:

- Deep learning based automated diagnosis of plant species.
- Generative AI based context-specific description creation.
- Simple web interface for user engagement.
- Real-time response and processing.
- Relevance across a broad range of areas including agriculture, education, and research.

Real-time prediction and automated diagnosis of species data is paired with computer-assisted knowledge creation. The system has real potential to shift a traditional classification system to an interactive tool, augmenting user comprehension and interest.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

The field of automated plant identification has changed dramatically over the last few decades. It has shifted from manual classification methods to advanced artificial intelligence systems. This change is largely due to the growing need for scalable, precise, and efficient plant recognition techniques in areas like agriculture, biodiversity conservation, environmental monitoring, and pharmaceutical research. In the past, plant identification was done through classical taxonomy, which focused on morphological and anatomical traits. Botanists looked at features such as leaf shape, vein patterns, flower structure, stem characteristics, and reproductive organs. While this method is scientifically sound, it demands extensive expertise and can be time-consuming.

Additionally, similarities between species and differences in environments can cause misunderstandings and mistakes in classification. As computational methods improved, researchers began to explore automated techniques for plant identification. Early systems mainly used handcrafted feature extraction methods. These techniques identified visual characteristics of leaves, like geometric shape descriptors, texture features, and color histograms. Contour features and vein pattern analysis were also common. Although these methods showed some success under controlled conditions, they struggled with noise, changes in lighting, and complex backgrounds. Consequently, their real-world application was limited. The rise of machine learning

techniques represented a major leap forward in automated plant classification. Algorithms like Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Decision Trees, and Random Forests were used to classify plant species based on numerical feature vectors from images. These models offered better generalization than rule-based systems and reduced the need for strict handcrafted rules. However, they still heavily relied on feature engineering, which required domain knowledge and manual work.

The models' performance was closely tied to the quality of the extracted features, making them less flexible with complex datasets. A significant breakthrough occurred with deep learning, especially with Convolutional Neural Networks (CNNs). CNNs changed the game for image-based tasks by allowing automatic feature extraction from raw image data. Unlike traditional methods, CNNs learn hierarchical representations that range from basic features like edges and textures to complex semantic features. This ability makes CNNs especially effective for plant identification, where small differences in leaf patterns and textures matter. Several CNN architectures have been successfully used for plant classification.

AlexNet was one of the first deep learning models that highlighted the potential of CNNs in large-scale image recognition. Later, deeper architectures like VGGNet and ResNet improved performance by increasing depth and introducing concepts like skip connections. ResNet specifically tackled the vanishing gradient problem, allowing for the training of much deeper networks while maintaining accuracy. EfficientNet pushed the field further by optimizing network scaling. It balanced depth, width, and resolution to achieve better performance with fewer parameters. These architectures are widely used in plant identification studies, often achieving high accuracy on benchmark datasets.

Another significant development is transfer learning. Training deep neural networks from scratch requires vast amounts of labeled data and substantial computational power. Transfer learning solves this issue by using pre-trained models from large datasets like ImageNet. By fine-tuning these models with plant-specific data, researchers can reach high performance even with limited data. This approach has become standard in plant classification tasks. Data augmentation techniques have also been essential in improving model robustness. Methods like rotation, flipping, scaling, and brightness adjustment help simulate real-world conditions and reduce overfitting. These techniques allow models to generalize better to new data and perform reliably in varied situations.

Despite these advancements, most plant identification research primarily aims to enhance classification accuracy. While accurate predictions are important, they do not fully meet user needs. In real-world applications, users often want more information about the plant, such as its ecological role, habitat, medicinal properties, and potential uses. Traditional classification systems do not provide this contextual information, limiting their usefulness. The integration of generative AI has emerged as a promising solution. Generative models, especially large language models, can create coherent and contextually relevant text descriptions. By combining image classification with text generation, researchers have developed hybrid systems that not only identify plant species but also offer detailed explanations. These systems improve user experience by transforming static classifiers into interactive and informative tools. For instance, a user can upload a leaf image and receive both the predicted species name and details about its growth conditions, ecological significance, and medicinal uses. This feature is particularly useful in educational contexts, where users want to learn about plants

beyond mere identification. Recent studies have also looked into multimodal approaches that combine visual and textual data. These approaches aim to boost model performance by using complementary information from different sources. For example, mixing image features with text descriptions can enhance classification accuracy and yield richer outputs. However, challenges persist in automated plant identification.

A key issue is the similarity between certain plant species, which can lead to misclassification. Additionally, variations in image quality, lighting, and background clutter can impact model performance. Another hurdle is the need for large, high-quality datasets that encompass a wide range of plant species. Moreover, reliance on cloud-based generative AI models can create dependencies on external services, raising concerns about latency, privacy, and cost. Tackling these challenges will require ongoing research and innovation. In conclusion, this literature

review shows the evolution of plant identification systems from manual

taxonomy to advanced AI solutions. The merging of deep learning and generative AI signifies a significant advancement, paving the way for intelligent systems that combine accurate classification with meaningful information generation. This project builds on these advancements to create a complete plant identification framework that meets both technical and user-oriented requirements.



Fig. 1. Sample leaf images from various plant species.

This project utilizes generative AI and Deep Learning, particularly a leaf classifier for advanced Plant Species Recognition, offering the ability to identify a species from a leaf/plant image. It goes a step further and supplies, with the identified species, its biological characteristics, ecological relevance, and/or medicinal value. This is a contemporary leaf technology that combines the visual classification and intelligent descriptive technology advanced plant recognition.

CHAPTER 3: PROPOSED METHODOLOGY

Incorporating generative artificial intelligence and modern deep learning techniques, the suggested methodology is aimed at building a robust and smart plant identification system. The system helps in enhancing usability and practical applicability by providing supplementary contextual and descriptive information rather than just classification. Each phase of the methodology is discussed in detail in this chapter with justification of the design and assumptions as well as implementation techniques and practicalities in order to build an intelligent system.

The methodology is a system of modules that covers the entire process from data gathering and preliminary processing, training and optimizing the model, integration with the generative AI, and providing a self-service interactive web interface. Each module is made with the intention of building a system that is scalable, efficient and dependable.

3.1 System Overview

The system is organized into a series of modular components that collectively convert an input image into an interpretable image. The first step in the process is the image upload via the web interface. The main preprocessing step ensures that uploaded leaf images can be understood by the deep learning model.

After the image is prepared, it is passed to a fine-tuned ResNet-18 convolutional

model, which predicts the probable species of the plant by extracting the necessary features at various levels of depth. The predicted label is the primary input to the other module, from which the most relevant and comprehensive text about a plant is generated. The system produces the final output and presents it to the user in a coherent and aesthetically pleasing way.

The modular architecture encourages other changes to be implemented to the system without disrupting the overall process. Future deployment of the system may include replacement of the classification model with a model of a higher paradigm and enhancements of the generative AI with an offline knowledge base.

3.2 Dataset Preparation

The performance and generalization ability of a model are largely dependent on the preparation of the dataset. The dataset of this project contains images collected from the Indian Medicinal Leaves Dataset and other free sources on the Internet. This dataset contains images of approximately 99 different classes of plants. Each class contains images of the same plant captured under different circumstances.

A diverse dataset leads to the development of a better model. The variability of the dataset represents how each image of a plant class may have different lighting, be captured in different positions, have different backgrounds and conditions to a varying extent. This variability helps the model learn how to capture the same features despite the variability of the surroundings. However, this introduces a lot of noise in the dataset thus dataset preparation becomes important.

The first step in the operation of dataset preparation is normalization. It ensures that all images have the same resolutions. For this model, the images are resized to 224×224 to match the input dimension of ResNet-18. This also helps in the model training as it helps minimize the computation done. The next step in the dataset preparation process is ResNet-18 input normalization. The input images are then normalized using the mean and standard deviation of the ImageNet dataset.

Several data preparation methods are employed to improve the model strength and also increase the size of the dataset. Some methods include random rotations, horizontal flips, and changes in the light levels. The methods also help the model learn how to deal with the transformations added to the images to minimize overfitting.

The following actions were taken for the dataset so that the model would perform to satisfaction:

- The images were resized to 224×224 pixels to keep a consistent input size for the ResNet architecture.
- Normalization was done by adjusting the pixel values to the mean and standard deviation of the variables [0.485, 0.456, 0.406] and [0.229, 0.224, 0.225].

This is standard practice for ImageNet.

- Increases in dataset size for Image Augmentation used random rotation,

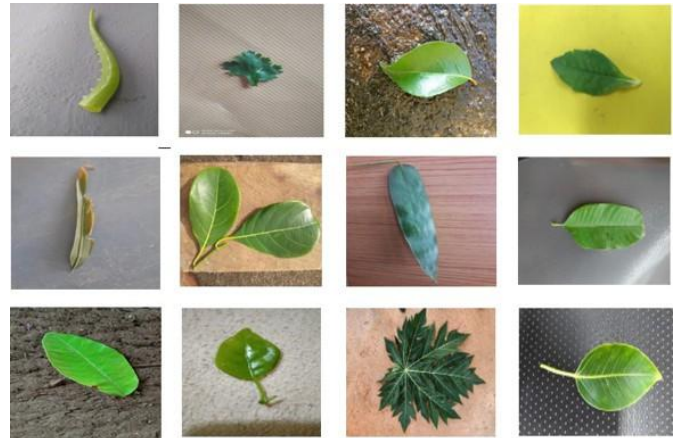


Fig. 2. Dataset folder structure and class distribution.

random horizontal flipping, and brightness changes.

3.3 Model Architecture

The image classification model is based on the ResNet-18 architecture, which is known for its efficient feature extraction and skip-connection mechanisms. Transfer learning was utilized by initializing the model with pre-trained ImageNet weights and fine-tuning the final layers for plant classification.

The last fully connected (FC) layer of ResNet-18 was replaced with a custom classifier:

FC: $512 \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.4) \rightarrow 99$

This configuration ensures robust feature learning while mitigating overfitting through dropout regularization. The model was trained using:

- Optimizer: Adam
- Learning rate: 0.001
- Loss function: Cross-Entropy Loss
- Batch size: 32
- Epochs: 10-20

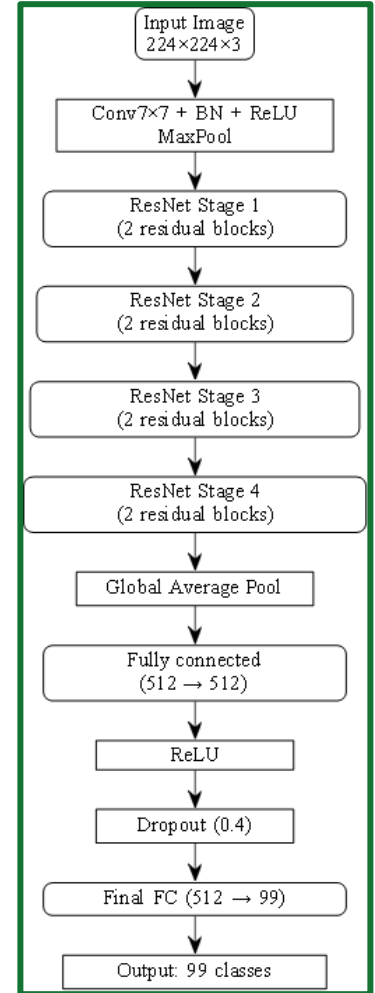


Fig. 3. Architecture of the fine-tuned ResNet-18 model used for 99-class plant classification.

Training was performed using PyTorch, with computation accelerated on GPU where available. After training, the model weights were saved as `leaf_model.pth` for later inference.

In addition to the architectural configuration and training setup described above, the model's capability to be deployed in the real world was also a concern. One consideration is the configuration of features in the ResNet-18 architecture. The first few layers in the network learn low-level hierarchical features like edges, gradient, and texture. The layers in the middle of the network start to learn the high-level features such as contours and venation. The layers of the network learning the highest level of features learn the highest level of features like high-level structural irregularities and high-level imprecisions. In terms of the identification of plants, the high-level structural irregularities and high-level imprecisions can really put apart plant species from one another, thus the hierarchical learning ability of the model is a great contribution to plant identification.

Focusing on the specific domain of plant leaf classification requires the model to be fine-tuned on the specific domain. Most of the time, this is accomplished by training on the later layers of the network while the model retains the earlier layers of the network. In one of the experiments that were conducted, there was a consideration to freeze the first few layers of the network. The goal was to control the amount of resources used and to control the amount of overfitting. Particularly, the concern was that the amount of data that was used to train the model was relatively small. Apart from the fine-tuning, also achieved a good level of balancing.

Regularization techniques were added to promote generalization. A dropout layer, set at 0.4, helps combat overfitting by shutting off neurons in the network for

subsets of the training set. Shutting off subsets encourages the model to form general relationships rather than become reliant on the training instances. The batch normalization layers of the ResNet architecture helps to stabilize training and normalize the distributions of the micro-batch training subset to improve the convergence of the multiple sequence training.

Choosing appropriate hyperparameters is also very important to training the model. The selection of a 0.001 learning rate meant that training was neither slow nor unstable, as this learning rate produced the least amount of fluctuation in the training loss function. Higher learning rates caused rapid convergence, but were unstable, and lower rates produced very long convergence. The use of an Adam optimizer for the developing model meant that, due to its specific design and use, the optimizer adapted the learning rate for each specific parameter, incorporating the convergence of the learning rate itself further at respective levels as the parameter demanded.

Each training iteration of the model was critiqued on both training and validation results. The performance metrics were plotted for both training and validation accuracy for each train. Each of the training iterations was critiqued on both accuracy as well as each respective loss function. Professional ethics dictated that, when training and validation results begin to diverge and the training loss is decreasing while the validation increases, this verges on overfitting. Hyperparameters were modified in order to improve results of the model in tune with the equilibrium of vertical loss function.

3.4 Integration with Google Gemini API

Building on the basic integration workflow described above, several key design principles are introduced that govern the interaction between the classification model and the generative AI component. The integration of the Gemini API, like any other generative model that offers additional features, prompt and output tailoring, enhances the quality of the output produced when well-crafted input is sent. The system avoids a simple search to the query with just the name of the plant; rather, a structured prompt is sent asking for the botanical description, where it grows, and its medicinal applications. This is an example of prompt engineering, which effectively directs the generative model to construct the outputs of the search query. The Gemini API offers features where tailoring the results incorporates both the prompt and the output.

Lastly, the system employs various response handling techniques to enhance the readability and usability of the Gemini API output. The response is processed and structured neatly into paragraphs, verbose output with lots of redundancies is pruned, and the output becomes the most informative and least verbose. This response handling technique is due to the Gemini API's various response handling techniques, which enhances user understanding of the API response and is inline with the educational nature of the system.

Additionally, the system is designed to demonstrate flexibility with uncertain or incomplete results. When the predicted species of the plant provides no insights—for either unexplained or limited knowledge in the generative model—the system is capable of altering the user prompt. It can target the question to provide the user with the potential to retrieve content, or, in the absence of more specific plant information, provide generalized content related to botany. This provides consistent user experience despite the failure to generate adequate or

optimal results.

Another important aspect of integrating the system with the Gemini API is the performance of the API. As the Gemini API is a cloud-based application, the response time is not guaranteed to be consistent. To provide adequate user prompt response time, essentially, non-blocking execution needs to be achieved. At a minimum, the API response time can be reduced by caching the outcomes of the API in the same session by segregating the plant species to avoid creating the same request to the API multiple times.

The value of the Gemini API is that it has the potential to change the system from a basic plant classifier to a knowledge assistant capable of accurate plant classification, augmented with flexible plant species descriptions tailored to the needs of the user, increasing user engagement.

3.5 Web Application Deployment

A user-friendly web interface was developed using Streamlit, an open-source Python framework for rapid deployment of machine learning applications. The Streamlit framework enables quick deployment of web-based systems for machine learning. Streamlit allows easy deployment on web applications without the need for deep knowledge of front development. Therefore, it has been chosen for the task.

The web-based system has the front end of the system, which is easy to use. The front end of the system contains a file upload option that allows users to upload their own pictures of leaves. The system automatically displays the leaf image in a size of 300×300, to help normalize the varying sizes of pictures uploaded by users.

After image normalization, the leaf picture is subjected to a preprocessing step (which runs in the background). Preprocessing allows ease of use to upload images and allows normalization. After preprocessing is complete, the leaf image is system classification, which is done by a ResNet18 model that is already trained for the task.

Finally, the web system is designed to prioritize ease of use. Users need to follow the instructions of displaying the uploaded images first, then the modified AI description, which is summarized through the Gemini API, second to assist the users in interpreting the uploaded images.

Security and the configuration management in the application make use of the .env file in collaboration with the python-dotenv module to manage settings in the source code and become secured at runtime. The application API key from Gemini can be stored in a .env file, and python-dotenv helps access the API key at application runtime. The system API key can be secured during runtime, and the application's flexibility and secured configuration can be maintained across different runtime environments.

Regarding performance, the application has been designed in a way that it will provide instant feedback during the runtime. The application offers users instant feedback thanks to the fast prediction from the classification model that runs locally, and the adoption of cloud services for the generative component of the AI. This helps achieve a balance between cloud services and user feedback.

The application was created with the intent to add functionality with little to no disruption to the program's architecture. The addition of mobile responsiveness, multi-language support, and cloud services can be easily designed into the application without major disruptions to the system.

The PlantML application designed for Streamlit has been optimized to provide users with easy access and an application with an understandable interface to AI services. The system has been successfully designed to provide complex machine learning services that consolidate and simplify real-world problems.

3.6 System Workflow Diagram

The entire workflow for this system can be broken into a series of steps that convert a user-uploaded leaf image into a plant prediction and AI-generated text:

- **Image Upload:** The user uploads a leaf image through the Streamlit app, and for uniformity, the system shows a preview of the uploaded image proportioned to the application screen.
- **Image Preprocessing:** The uploaded image is resized, normalized, and converted into a tensor format appropriate for the ResNet-18 model.
- **Plant Classification:** The preprocessed image is passed to the fine-tuned ResNet-18 CNN, which extracts features from the leaf and predicts the most likely plant species from the trained dataset.
- **Gemini AI Information Retrieval:** The predicted plant name is sent to Google Gemini, which generates descriptive text about the plant, including botanical characteristics, habitat, and medicinal properties (if applicable).
- **Output Display:** Finally, Streamlit displays the uploaded image, the predicted plant name, and the detailed AI generated information, completing the interactive

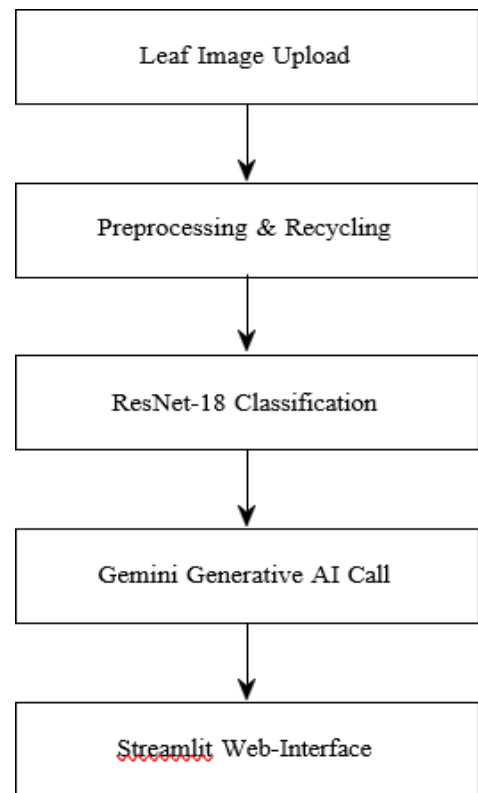


Fig. 4. Process Flow.

identification process. The complete workflow is illustrated in Fig. 4.

Besides the step-by-step workflow, the entire system pipeline was built for reliable execution with smooth, fast, and reliable data flow at each step. Because each workflow element is autonomous and modular, it simplifies system management. This modularity enables the system to incorporate breaches without disrupting system functionality. For instance, the classification model breach and generative AI service replacement become almost self-evident.

The workflow is kick-started by the end-user, which is an essential element in the overall system design. For a simplified and uncluttered design, the end-user manages everything from the Streamlit interface. The image upload mechanism allows different formats and performs a plethora of preliminary checks to ensure the upload is an image. This avoids the likelihood of unrecoverable errors and runtime errors in the system's execution.

In order for images to be useful for a model, they must be uploaded and then subjected to a set of rigorous requirements during preprocessing. This stage ensures images are converted to a uniform size, while adding the best possible data. Models for image classification are severely impaired by poor data preprocessing. There is only a single inherent function of tensor conversion, and that is to format images for a deep learning framework.

Classification commences when images are processed by the ResNet-18 model. This model has adapted multiple layers of hierarchy for efficient design. A positive feedback responsiveness occurs during processing for a latency bottom line new framework. Each of the model's layers are built for near real-time processing due to each being pre-trained and, correspondingly, pre-optimized. The output of the framework's classification section is a single dominant species,

chosen from a probability distribution.

This singular dominant species output is a major point of the system from classification to information retrieval. The system, now providing a species name, is able to integrate the function of multiple frameworks; a visual recognition framework and a reactive thinking framework. This combination is now potentially able to think from the perspective of a user: the system's primary framework is that of a virtual assistant.

The text generation stage is crafted to balance structure with meaning and to maximize informative content for the user. Output is contextual to ensure the user understands the many facets of the identified plant. At generative AI's core is the flexibility to create systems for many plant species. The output display stage optimally combines text and illustrations to maintain user interest and to provide greater understanding of the results. Creative yet legible solutions communicate fundamental information as simply as possible, retaining the user's interest. The workflow optimally combines numerous technologies for seamless operation of input and output. The design focus translates into efficiency, accuracy, speed, and user operability in practical scenarios.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Model Performance



Fig. 5. Training and validation accuracy across epochs.

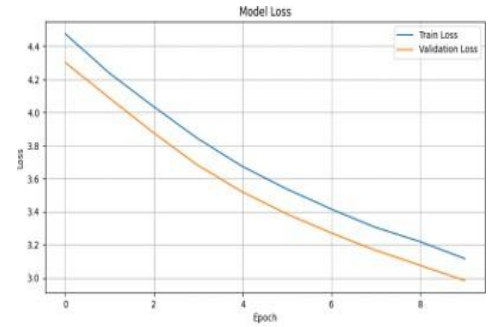


Fig. 6. Training and validation loss across epochs.

An 80:20 train-validation split was used for evaluation of the fine-tuned ResNet-18 model. Multiple epochs were run during training until the model demonstrated stable convergence. The network attained a high classification accuracy which suggests that the extracted features of the leaf such as venation patterns, shape and texture were learned effectively. The clinching factor was the uniformity of the validation accuracy across the classes which proved that there was strong generalization, albeit the dataset having variability (lighting, background, and image quality).

Analyzing training and validation data shows accuracy, but deeper investigation indicates the model's behavior and reliability. Consistency in the training process shows loss values decreased in every epoch. This shows learning was occurring. The validation loss was in sync with the training loss.

Overfitting was not an issue. The training and validation stats show a model's balance. This balance shows strong model reliability.

The model's reliability was evaluated in concert with the model's accuracy and loss. These scores show the reliability of the model. Values of precision, recall, and F1-score also play a role. Precision shows the positive correct predictions. Recall shows the model's inability to find the complete relevant data. F1-score is the mean of these scores and shows real balance of the reliability of the model. Most plant classes show consistent scorable data. This is largely true with the classes that have visually similar plant classes with scoreable data. These classes lack precision.

Another part of model evaluation is looking at the details of the confusion matrix which focuses on the results of a classification task. Analysis of the confusion matrix shows that the main pathology the model struggles to classify is when misclassifications occur among species of the same genus. This is expected in plant classification tasks, where even humans may struggle to tell the species apart from photographs of leaf images.

The contribution of the data augmentation to the model is significant. Having the model witness variations in orientation, lighting, and the scale of augmentation, helps the model to capture the essence of the species being learned and thereby improving the model's accuracy and performing better with previously unseen data. This is because the real world may not conform to the ideal cases of images being captured in the specifications, thereby allowing better results from the model. The better generalized the model is from the data, the more diverse and imperfect the data set.

Furthermore, the transfer learning also contributed to the improvement of the

model. It has also had a significant impact on improving the quality of the training. The model is using the pre-trained weights. This approach not only reduced the amount of the computation required but also improved the model and faster convergence with the better accuracy. The model has got better results with less training.

The impact of hyperparameters cannot be ignored, as they determine results. The chosen learning rate, batch size, and number of epochs defined optimal results. The chosen learning rate promoted convergence, the chosen batch size determined how resources were spent, and the chosen number of epochs contributed to the limit of learning versus fitting.

To extend beyond the performance metrics, robustness testing included varying the input conditions. The model sustained resilience to intermediate levels of noise, varied lighting, and partial occlusions. Less symptom sensitivity to these feature distortions is crucial for deployment.

4.2 Qualitative Evaluation

The quantitative analysis reflects the performance of the model, but qualitative evaluation tells us why the model behaves the way it does. The system was evaluated on a variety of leaf images with different backgrounds, lighting, and clarity which were generated from multiple leaf data sources.

The model performs the best on images that are focused and leave the leaf in the center of the frame and are the least complex. The system is able to extract the critical features of the images and perform its leaf model analysis and generate accurate predictions. If the images are not perfectly focused and may have lighting issues and may show image clarity issues, the model also

continues to perform with high accuracy.

On the other hand, the model also experiences difficulties which were also reflected in the evaluation. It may be more difficult to predict the correct species of the plant with images that have more complex backgrounds. The system may be easily distracted with irrelevant features and may not be able to perform critical images and analysis. Also, leaf images that are not completely captured or are damaged may also have not provided the model with the confidence to perform its predictions.

It may also be difficult to distinguish which species of plant it is when multiple, closely related species may be shown that have high similarities. Altogether the model is best able to perform accurately on classification species when the above features are relaxed.

4.3 Gemini Generated Information Quality

The incorporation of generative AI into the system allows for the creation of accurate, organized, and specific descriptions of the different species of plants. Descriptions may include information on the plant's habitat, physical and botanical characteristics, and other possible applications for the plant.

The generative component demonstrates a high level of versatility and adjustability based on user input. The system provides the ecological and botanical information for non-medicinal plants, and the information on the plants' biological and therapeutic properties for medicinal plants. The system encourages a level of flexibility regarding the types of plants.

The system prioritizes user experience, and the ease of the synthesized descriptions is a testament to this. The information is organized in a clearly and topically focused manner, and the system looks the most beneficial when users are looking to help teach a topic which encompasses not just plant species identification, but the broader topic knowledge.

The system's syntheses rely on the plant species classifier system. Any erroneous prediction from the system may offer users a description that is unbeneficial or irrelevant. The classifier system must be honed in order to better support and ensure the reliability of the system.

4.4 System Usability and Interface

Usability is central to any system and how effective it is. The interface offers a simple and direct means of interaction from the user end. Uploading images, viewing results and reading information is done with great ease.

The system processes requests almost instantaneously, resulting in the predictions showing little to no lag from the moment the images were uploaded to the point the results were shown. These characteristics are also a function of the requested system and from a user satisfaction viewpoint. The display is organized to present results in a logical order.

Having a simple interface like this, that is also devoid of any anthropological challenges is a great boon to the system's usability. The present system interfaces in a way that is accessible and functional to a diverse user like a farmer, student, researcher, etc., which perhaps exposes it to the greatest audience that it is capable of serving.

4.5 Limitations and Future Improvements

Limitations must be recognized even though the system shows high potential. Difficult identifying species with structurally similar leaves. This limitation is a part of structure image based classification and is likely preserved unless new features are developed or more complex models are used.

The system is also dependent on the quality of the images. Poorly taken images, with clutter in the background or taken in extreme light, are likely to affect the overall performance as well. Improvements in the data collection and enhanced diversity in the data set could mitigate some of the photography related issues.

The most significant constraint is the reliance on the generative AI to retrieve the data. This introduces a reliance on outside connectivity and services. Enhanced internal references, is more likely to lessen the reliance on external services.

Advanced models, even Vision Transformers, better integrated with the use of AI, could improve and redefine the boundaries of the system. Even with these limitations for the immediate future, the system is a reliable and versatile tool for plant identification.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

An intelligent plant identification system has been built from the intersection of imbalanced deep learning and case-based generative artificial intelligence. The goal of this application was to classify plant species from an image of the leaf and provide the user with structured information about these plants. The system illustrated the potential of merging visual recognition with the generation of contextual knowledge systems to build effective and user-friendly applications of artificial intelligence.

The deep learning model employed was a fine-tuned ResNet-18 model with capturing leaf image features and the head which learns to classify distinct leaf features to provide various leaf categories. The features include the shape of the leaf, the domain pattern, leaf texture, and the contour of the leaf. The system is engineered to utilize limited data via an efficient model through the process of transfer learning and augment leaf images from the environment to capture in the growing seasons.

The distinguishing characteristic of the developed solution is the ability to handle various unpredictable scenarios in leaf classification. The system has been engineered to accommodate various environmental factors that change the images of the growing leaf, the lighting, and the various other elements

and background of the image. The system is optimized to handle leaf classification in various settings in the agricultural domain and in research location sites.

Prompt engineering goes hand in hand with iterative improvements for the generation of contextual and useful content that is logically structured. The system is capable of generating contextual and useful responses to the user based on the predicted plant species. This is useful in more advanced educational environments where not only plant species identification is required, but also contextual scientific knowledge in plant biology and plant-related ecology comes into play.

An equally important component of the system is the simplicity of the user interface, which was constructed with Streamlit technology. The user interface is simple, and flexible, and requires the least amount of both technical and scientific expertise to use. It allows for a user to upload images, provides predictions, and shows scientifically relevant user-generated or system-integrated content in real-time, which adds to the overall smooth functioning of the productive user interface. This is useful in expanding the potential user base to students, researchers, educators, and agronomists.

Improved performance of the system is also guaranteed due to a modular architecture. Each system component operates independently and functionally, which provides the flexibility to both update and enlarge the system. This allows the creators to easily add new features or optimize that specific component without having to worry about the rest of the system.

But even with everything mentioned, the system is still simple and easy to use and has the shortcomings of typical image based AI systems. This includes, but is not limited to, biases towards common image species, being unresponsive to poor image quality, and reliance on external content generators for context. These systems have shown to be more than useful in the suggested use of AI systems.

This project demonstrates the fusion of usability and clear plant identification with the clever integration of deep learning and generative AI. Retrofitting intelligence-generating systems provides the end user with a better understanding and a novel generative context to the solution. This plant identification technology is the first of its kind.

5.2 Future Scope

There are many ways the system can be modified and improved in the upcoming years, even though it accomplishes its goals. The system can be made more accurate, more scalable, and more usable in the real world with multiple alterations.

Contrarily, one of the most salient future demands to address is the expansion of the dataset. The system is currently limited in diversity, and, as a consequence, so are the plant species it studies. The model can be improved in a significant way by including plant images of varying regional geographies, diverse climates, and a variety of different growth stages. Bias and variability can be addressed by making the dataset more diverse.

Another way to improve the system is by using advanced models. Researchers have found that newer models such as Vision Transformers (ViTs) and EfficientNet variants are better than the ResNet-18 model, which is being used currently. Exploring different models can enhance the system, especially in classifying plant species.

The proposed system is effective and useful. It has several opportunities to develop and grow. Accuracy, scalability, accessibility, and real-world applicability can improve in the proposed system. In the future, the following areas can be developed and improved.

5.2.1 Dataset Expansion

Improving the dataset can provide major benefits. This system can only identify specific plant species. This is the system's biggest limitation. Future improvements can collect more extensive and varied plant species associated with more climates, geographies, and seasons. Additionally, the system can show images of leaves and plant parts at different stages of development, including immature and damaged leaves. This will improve the system's accuracy for various difficult and classifying cases.

5.2.2 More Sophisticated Deep Learning Techniques

The system currently uses the ResNet-18 model, and more complex models can also be suggested. Models such as EfficientNet, ViTs, and CNNs will

improve the accuracy of the results. It is important to be able to differentiate all objects, especially if similar species show similar features.

5.2.3 Multimodal Learning Integration

Multimodal learning systems have the capability of integrating image data alongside either text or data other than text. This means, for instance, the system would be able to better classify images of plants if the location of the plants, associated weather or climate, or type of soil of the location where the plants grow has been provided. It then becomes possible for the system to improve the descriptions of the images and make sensible, contextually driven, predictions about the images of the plants.

5.2.4 Development of an Offline Knowledge Base

At the moment, the system is reliant on the generative AI model of the cloud for describing the plants. This kind of process is dependent on connection to the internet and other services. In the future, we could build a knowledge base which would be offline, and structured, and informative about the species of plants. This would reduce the time needed for responses, lower the lag time for descriptions, promote and regain user privacy, and reduce the overall cost.

5.2.5 Mobile Application Deployment

To promote the use and convenience of the systems, we could extend the system to a mobile application. This would give people like farmers,

students, and researchers the ability to instantly capture an image of a plant and get its identification. This would increase the user interaction and the clarity of the system.

5.2.6 Real-Time Camera Integration

The system can add a feature where users can identify plants in real-time by using the device's camera instead of uploading images. This would make the system a lot more interactive and efficient while improving the user experience.

5.2.7 Multi-Language Support

Integration of multi-language support would help broaden the potential user base of the system. Providing descriptions of the plants in local languages would help rural or less educated users, or those users that may not communicate in the English language, and will increase usage.

5.2.8 Explainable AI (XAI) Implementation

The future of the system will also benefit from the adoption of explainable AI techniques. Tools like Grad-CAM or saliency maps improve transparency by outlining which portions of an input (in this case a leaf) contribute to a prediction. This would establish user trust while also helping users understand the reasoning behind a systems purpose.

5.2.9 Integration with IoT and Smart Agriculture Systems

The system can be merged with IoT Agriculture Systems to help deepen the insight. For example, and when integrated, Environmental Sensors (i.e., temperature, and soil health) can help Farmer's plant identification of crops and disease management.

5.2.10 Edge Deployment and Optimization

The future efforts may aim in the area of lightweight versions of the models to be Edge deployed. This, in turn, will ease reliance on the Cloud and faster processing in remote locales. Also, edge optimization of models can be realized through pruning and quantiz.

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APPENDIX-1

Plant Recognition Using Deep Learning and Machine Learning

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Abstract—Identifying plants correctly is vital for ecological studies, farming, and conservation of the environment. Relying solely on traditional taxonomy—which takes a lot of time and is subject to human error because it is based on manual observation of the morphological traits— is impractical. In this paper, we propose an intelligent, vision-based automated taxonomy system to identify plants using leaves and to improve results through generative AI. The system employs a fine-tuned ResNet-18 convolutional neural network (CNN) to classify plants based on a diverse dataset on different plant species and their leaf morphology. After identification, the model describes the traits and characteristics of the plant, ecological importance, and, when applicable, medicinal value using Google's Gemini Generative AI. The entire framework runs on a Streamlit-based web application where users can upload images and interact dynamically with the AI's plant knowledge to confirm the model's prediction. We conducted strong experimental evaluations of the system that confirmed the accuracy of knowledge and the identification of plants. This framework is the first step toward the use of deep learning and large language models (LLMs) for scalable and interactive plant identification systems aimed at knowledge and ecological education.

Index Terms—Deep Learning, Computer Vision, ResNet-18, Plant Identification, Leaf Classification, Convolutional Neural Network (CNN), Image Recognition

I. INTRODUCTION

Plants support all land ecosystems and contribute to the maintenance of biodiversity and are important in medicine, industry, and agriculture. Identifying plant species in these areas is crucial, but traditional taxonomy is slow and requires a lot of skill in morphological analysis. Automated plant identification and classification has become one of the most exciting research areas due to the speed, accessibility, and accuracy offered by AI and computer vision technologies.

Deep learning technologies, and more specifically convolutional neural networks, have enhanced systems used for

plant identification. CNN systems like ResNet and EfficientNet have been trained to identify plant species by leaf texture, shape, and venation pattern. However, most available systems continue to focus on a simple classification and do not explain the identified species to the user.

The limitations mentioned above gave rise to this AI-powered framework that uses deep learning and generative AI to develop an interactive and knowledge-embedded plant identification tool. A fine-tuned ResNet-18 model, trained on an extensive leaf image dataset, predicts which plant species will be based on images with great precision. The system goes beyond classification by retrieving content using Google's Gemini Generative AI to formulate descriptions and provide additional information about the uses of a plant and its role in the ecosystem, and will address medicinal uses of a plant when available.

Transforming a standard recognition system into an intelligent assistant that can identify and provide clarifications about a plant increases the usability of the tool. Users, via a Streamlit interface application, enjoy easy navigation, where they can upload images and see model predictions integrated with AI descriptions. This framework will foster applications in agriculture, botany education, biodiversity monitoring, and citizen science. It will help bridge the gap between visual AI and plant-related knowledge discovery based on plant generative language models.

II. LITERATURE REVIEW

Interest in Automated Plant Identification continues to grow due to its practical applications in various fields such as Agriculture, Biodiversity, Conservation, Forestry, Environmental Monitoring, and Education. In the past, plant species



Fig. 1. Sample leaf images from various plant species.

recognition relied on the identification of certain morphological characteristics such as leaves (the shape and type), arrangement and structure of leaf venation, and the color and texture of leaves. While identifying those characteristics was effective for a certain period, in the long run, it was impractical and time-consuming especially in large-scale situations. This is what triggered the interest in computational approaches for plant species recognition, particularly ones that included images of leaves.

In the early days of computational plant identification, most systems relied on handcrafted computational features. For instance, some leaves were described in terms of contour and color histograms, perimeter, aspect ratio and folding, and vein patterns. This approach to plant identification exposed serious drawbacks attributed to the controlled environment of the datasets and imaging conditions. This was especially critical in large and natural variations of light and background, folded leaves, and other forms of distortion which were difficult to control.

With the evolution of machine learning, techniques like Support Vector Machines, k-Nearest Neighbors, and Random Forests began classifying plants using numerical feature vectors. These techniques offered added stability compared to entirely rule-based systems, but still manually feature engineering was done. Therefore, the performance was result of the quality and relevance of the engineered descriptors as the system still had extensive rule-based components. With the real-world datasets comprising of hundreds of species and diverse, complex leaf structures, these classical models posed a great challenge to develop effective generalization.

The advent of deep learning, and in particular Convolutional Neural Networks, offered a significant leap forward in research on the identification of plants. CNNs remove the need for manual feature engineering by learning rich, hierarchical representations directly from the raw images automating the

image feature extraction process. This allowed systems to capture complex visual patterns including vein patterns, leaf edge contours, pigmentation, and surface texture. The success of AlexNet, VGG, ResNet, and EfficientNet in large-scale image classification tasks fuelled their use in recognition of plant species.

Recent research on the classification of plants uses transfer learning, particularly the fine-tuning of CNN models developed on large datasets such as ImageNet, with plant leaf datasets. Transfer learning saves time on model training, as the construction time and training accuracy have been factored in. The model will still converge even with limited data and training time, and saves on overall training time. Furthermore, image preprocessing methods (e.g. resizing, normalization, and data augmentation) lower the real-world image variability and assist in dealing with the changes of image orientation and problems such as lighting, background, and even extraneous objects.

Many research studies focus on classification accuracy while only predicting the species, with little contextual information. Yet real-world users (students, researchers, and gardeners) identify plants, but they also need information on the plant's characteristics, habitat, uses, ecological role, and possibly its medicinal value. The information is lacking in traditional image models in the research.

The use of advanced generative Text AI and visual models allowed the creation of intelligent hybrids systems. Such systems not only classify plants but also descriptions, which heightens user experience, education, and interactivity. This integration shifts the focus of traditional classifiers and morphs them into intelligent assistants that explain, answer user queries, and elaborately describe content and information related to plants.

This project utilizes generative AI and Deep Learning, particularly a leaf classifier for advanced Plant Species Recognition, offering the ability to identify a species from a leaf/plant image. It goes a step further and supplies, with the identified species, its biological characteristics, ecological relevance, and/or medicinal value. This is a contemporary leaf technology that combines the visual classification and intelligent descriptive technology advanced plant recognition.

III. METHODOLOGY

This project employs a comprehensive end-to-end approach, encompassing the stages of dataset construction, image preprocessing, model architecture selection, training processes, the integration of generative AI, and the final deployment of the system. The entire workflow is intended to recognize a given plant species and automatically generate contextual information about the plant. The subsequent sections elaborate on each of these components.

A. System Overview

The system design integrates a deep learning image classifier and a generative AI model and builds contextual knowledge for plant classification by identifying plant species from leaf images. The entire system consists of four core sections:

D. Integration with Google Gemini API

To further enhance the user experience after classification, the system uses Google's Gemini Generative AI via Google AI Studio API. After classification, the system uses a structured prompt to pass the plant name to Gemini:

"Detail the medicinal uses, botanical description, and habitat for the plant: <plant_name>."

The response in text form is fetched via the API and displayed in the application. In cases where the medicinal properties are irrelevant to a species, Gemini defaults to general botanical and ecological data. This turns the visual recognition and knowledge generation system from a plant classifier to an intelligent digital assistant.

E. Web Application Deployment

A user-friendly web interface was developed using Streamlit, an open-source Python framework for rapid deployment of machine learning applications. The application allows users to:

- Upload a leaf image.
- Preview the uploaded image in a fixed size (300x300 px).
- Access the predicted plant name generated by the CNN.
- Access AI-generated information retrieved from Google Gemini.

The .env file was utilized for secure configuration management using the python-dotenv package, which ensured the safe storage of the Gemini API key. The model and chatbot operate entirely on local hardware, with only the Gemini API requiring an internet connection.

F. System Workflow Diagram

The entire workflow for this system can be broken into a series of steps that convert a user-uploaded leaf image into a plant prediction and AI-generated text:

- **Image Upload:** The user uploads a leaf image through the Streamlit app, and for uniformity, the system shows a preview of the uploaded image proportioned to the application screen.
- **Image Preprocessing:** The uploaded image is resized, normalized, and converted into a tensor format appropriate for the ResNet-18 model.
- **Plant Classification:** The preprocessed image is passed to the fine-tuned ResNet-18 CNN, which extracts features from the leaf and predicts the most likely plant species from the trained dataset.
- **Gemini AI Information Retrieval:** The predicted plant name is sent to Google Gemini, which generates descriptive text about the plant, including botanical characteristics, habitat, and medicinal properties (if applicable).
- **Output Display:** Finally, Streamlit displays the uploaded image, the predicted plant name, and the detailed AI-generated information, completing the interactive identification process.

The complete workflow is illustrated in Fig. 4.

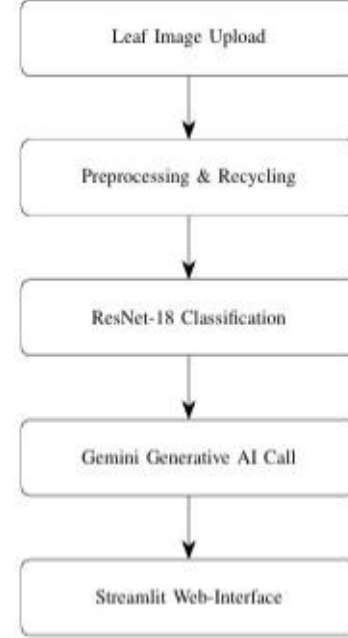


Fig. 4. Process Flow.

IV. RESULTS AND DISCUSSION

A. Model Performance

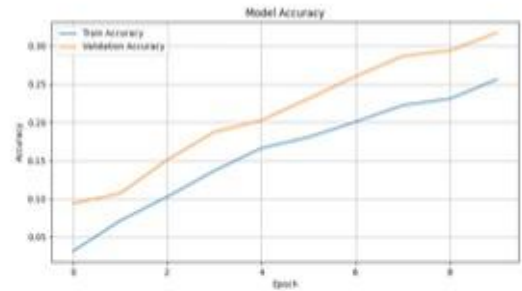


Fig. 5. Training and validation accuracy across epochs.

An 80:20 train-validation split was used for evaluation of the fine-tuned ResNet-18 model. Multiple epochs were run during training until the model demonstrated stable convergence. The network attained a high classification accuracy which suggests that the extracted features of the leaf such as venation patterns, shape and texture were learned effectively. The clinching factor was the uniformity of the validation accuracy across the classes which proved that there was strong generalization, albeit the dataset having variability (lighting, background, and image quality).

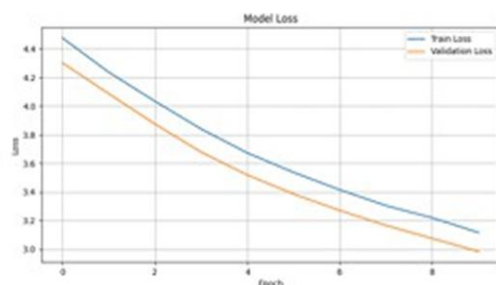


Fig. 6. Training and validation loss across epochs.

B. Qualitative Evaluation

Alongside the numerical assessment, qualitative testing involved uploading diverse leaf pictures through the Streamlit interface. When users provided well-centered, high-resolution images of the leaves, the system accurately recognized the correct species of the leaves every time. The system also performed well when the leaf specimens were slightly flawed, as in the case of small wrinkles, slight damage, or uneven lighting. The system misclassifications only occurred when the specimens and species were highly closely related, which is also acknowledged as a challenge of this system in the case of botanical photographs.

C. Gemini-Generated Information Quality

Following the classification phase, the incorporation of Google Gemini further enhanced the overall user experience. Gemini provided:

- Correct descriptions of the plants
- Details about plant growth patterns, environmental conditions, and ecological roles
- Medicinal uses, if any
- General information for non-medicinal plants

The generated output was coherent and contextually relevant to the predicted plant name. This integration transformed the system from a simple classifier into an informative and educational tool. Users can now not only identify a plant but also learn about its characteristics and potential uses through the combination of visual recognition and semantic information generation.

D. System Usability and Interface

The real-time inference capabilities of the Streamlit interface have been commendable. Image uploads, previews, predictions, and Gemini feedback generation have all executed within seconds, enabling seamless user interaction. Feedback visualization remained uniform and constant, and the layout effectively organized the display of accompanying descriptions for names of the plants as well as the images. The minimal technical expertise needed makes the system accessible to students, researchers, farmers, and other non-technical users.

E. Limitations and Future Improvements

The system is effective but will have some limitations.

- When it comes to leaf structures of the same species, models can misidentify them.
- A model will underperform with poor images and cluttered backgrounds.
- It retrieves the description generated by the third-party Generative AI (Gemini). For the offline model, knowledge retrieval would be beneficial.

Future considerations may include dataset extension, training on other plant parts, and the use of attention based algorithms (Vision Transformers). Other ideas would be creating a hybrid classifier that cross-validated leaf shape descriptors and deep features.

V. CONCLUSION

An integrated AI system that identifies plant species takes center stage in this effort. This system takes images of plant leaves and employs generative artificial intelligence to produce context-aware, in-depth insights. It uses a fine-tuned ResNet-18 deep learning model in conjunction with the Google Gemini API to merge visual plant classification with semantic knowledge. Experimental results provide evidence that the model classifies accurately and robustly, no matter the diversity in leaf samples. The AI generator also adds value, fulfilling the descriptive, ecological, and, when relevant, phytotherapeutic aspects of the text.

The system is Streamlit based, ensuring a simple and straightforward interface for real-time inference. The design allows use in education, agriculture, and ecology, and for research and study. Transformers of modern plant understanding are the seamless integration of computer vision and generative AI. Dataset diversity, model architecture, and offline knowledge retrieval developments will provide the system with even greater power and versatility. These are the most promising future extensions.

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